

Project Design Phase-II

Technology Stack

(Architecture & Stack)

Date	
Team ID	SWTID-2026-6582
Project Name	Smart Library Request Workflow in Service now
Maximum Mark	4 Marks

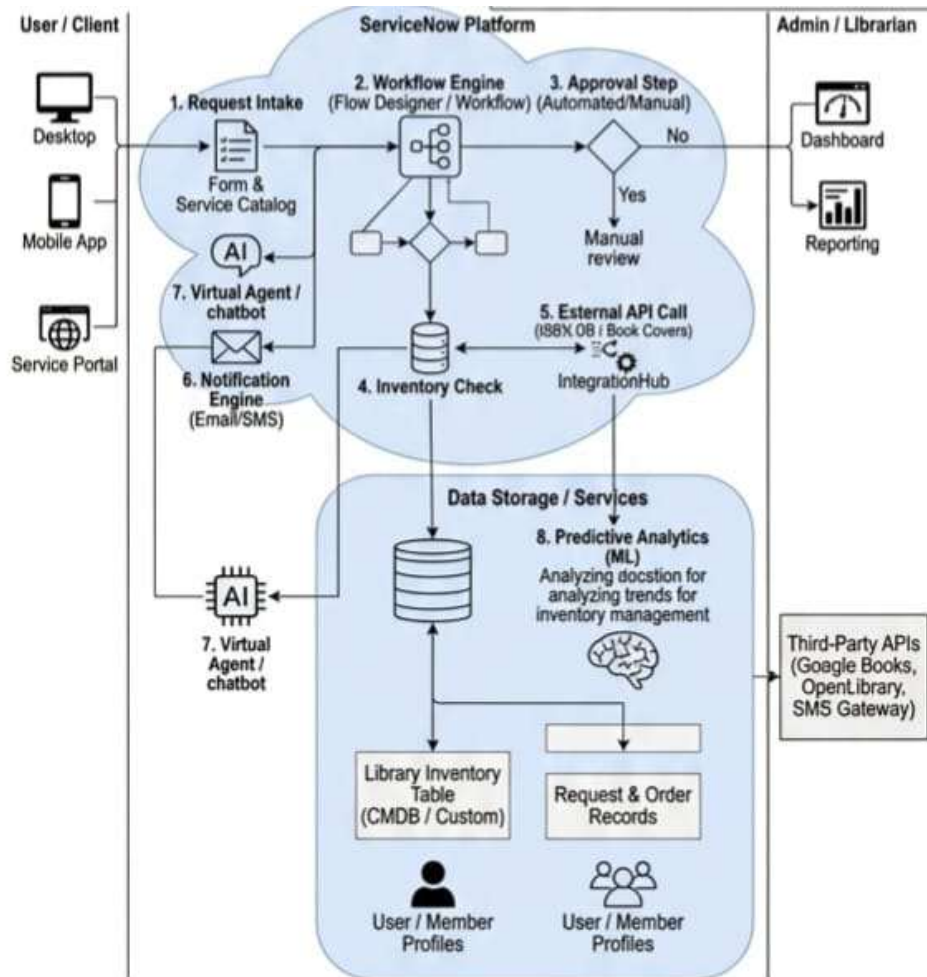
Technical Architecture:

Based on the architectural requirements for Project Design Phase-II, here is the Technology Stack and Architecture breakdown for your Smart Library Request Workflow in ServiceNow.

This structure follows the specific table formats shown in your reference image, focusing on how ServiceNow components act as the "engine" for the library system.

Reference:

<https://www.servicenow.com/workflows/it-workflows.html>



Guidelines:

- Include all the processes (As an application logic/Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

S NO	Technology Stack	Description	Technology
1	User Interface	How user interacts (Service Portal, Now Mobile App, Virtual Agent)	HTML, CSS, JavaScript/AngularJS (Service Portal) /React (Now Experience)
2	Application Logic-1 (Request Flow)	Core workflow logic for requests and approvals	ServiceNow Flow Designer/Workflow
3	Application Logic-2 (Automation)	Server-side automation for due dates and inventory updates	ServiceNow Business Rules / Script Includes (JavaScript)
4	External Interfaces (Integrations)	Connects to external book databases and notification systems	ServiceNow Integration Hub (Rest/SOAP spokes), MID Server (if required)
5	Machine Learning (AI)	Predicting demand and recommending titles	ServiceNow Predictive Intelligence (ML Solution)

1. Solution Requirements (Functional & Non-functional):

2.

FR No	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication	• Login via Institutional ID (SSO). • User profile sync with library membership level. • Password recovery via Email/OTP.
FR-2	Resource Discovery	• Search catalog by ISBN, Title, or Author. • Filter by physical book, e-book, or journal.

		• View of real-time availability status
FR-3	Request Workflow	• Submit "Borrow"" request for physical items. • Instant download link for digital resources. • Automated waitlist management for checked-out items
FR-4	Inventory & Tracking	• Barcode/QR scan for check-in and check-out. • Automated due-date calculation. • Fine generation for overdue items.
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3. Technology Stack (Architecture & Stack):

This table maps the library system components to the specific ServiceNow technical stack.

S. No	Component	Description	Technology
1	User Interface	Student/Librarian portal and dashboards	Service Portal (AngularJS / Now Experience)
2	Application Logic-1	Core request/approval routing logic.	Flow Designer / Playbooks
3	Application Logic-2	Automated validation (e.g., checking loan limits).	Business Rules (JavaScript / ES12)
4	Application Logic-3	Conversational bot for book status.	Virtual Agent (NLU / GenAI)
5	Database	Metadata storage for books and users.	Raptor DB (HTAP) / MariaDB.

6	Cloud Database	Storage for large digital assets (e.g., e-books).	AWS S3 / ServiceNow Managed Storage
7	File Storage	Handling PDF previews and attachments.	ServiceNow Attachment Table (sys attachment)
8	External API-1	ISBN metadata and book cover fetching.	Google Books API (REST via Integration Hub).
9	External API-2	Notification gateway for SMS alerts.	Twilio Spokes / Integration Hub
10	Machine Learning	Recommending books based on borrowing history.	Predictive Intelligence
11	Infrastructure	Global hosting and cloud runtime.	ServiceNow SaaS Platform (Multi-instance)

4. Application Characteristics:

S. No	Characteristics	Description	Technology
1	Open-Source Frameworks	UI and utility libraries are used in the portal.	AngularJS, Bootstrap, D3.js (for analytics)
2	Security Implementations	Access control and identity protection.	ServiceNow ACLs, OAuth 2.0, MFA.
3	Scalable Architecture	Tiered logic and data separation.	3-Tier Cloud Architecture (Presentation/Logic/Data)
4	Availability,	Continuity and disaster recovery.	Load Balancers, Multi-region Data Centers

5	Performance	Speed and efficiency optimization	Raptor DB, Glide Cache, CDN for UI assets
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4. Architectural Diagram Concept:

To follow the "Guidelines" in your third screenshot, your diagram should visualize:

- User/Admin Layer: Interacting via Web UI or Chatbot.
- ServiceNow Cloud Layer (Logic): Containing Flow Designer (Workflow), Business Rules (Validation), and Virtual Agent (Interface).
- Data Layer: Connecting to the internal Library Database and external APIs (like Google Books) for real-time metadata.

Reference:

- <https://www.servicenow.com/products/app-engine.html>
- <https://www.servicenow.com/products/platform-flow-designer.html>
- <https://www.servicenow.com/products/integration-hub.html>
- <https://www.servicenow.com/products/service-portal.html>